Set Vision Policy: Learned Perception to Browser Action

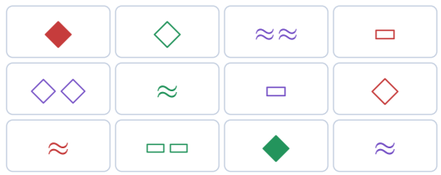
LOCAL VISION-TO-ACTION POLICY

41,918 PARAMS | NO ATTENTION

Compact CNN - four specialists - differentiable relation scorer - local WASM deployment

1 INPUT OBSERVATION

Live board screenshot



Screenshot — H x W x 4 RGBA

Cards — 12 clickable rectangles

2 DATA PREPROCESSING PIPELINE

Capture
visible browser viewport

→ **Crop**
12 DOM rectangles
geometry only

→ **Area average**
integer source bins
anti-alias stable

→ **Normalize**
RGB / 255
clamp 0..1

→ **Pack**
12 x 1,890
f32 -> WASM

SOURCE CROP
variable W x H x 4

CARD TENSOR
[3, 21, 30]

BOARD TENSOR
[12, 3, 21, 30]

CONTRACT
browser = trainer fixture

PROCEDURAL TRAIN SOURCE
Production Card renderer | four themes | multiple raster widths | seeded color, texture, geometry, and lighting variation

3 INTERMEDIATE REPRESENTATIONS



RGB card



shared feature map

Input	[N, 3, 21, 30]
Shared	[N, 12, 11, 15]
Per head	[N, 8, 6, 8]
Embedding	[N, 24]
Attr logits	[N, 4, 3]

BOARD CACHE
12 cards encoded once
144 attribute logits

6 TRAINING OBJECTIVES

SET LOSS
CE([0, score], y-set)

AUXILIARY LOSS
4 x mean CE(attribute logits, labels)

$$\mathcal{L} = \mathcal{L}_{\text{set}} + 4\mathcal{L}_{\text{attr}}$$

Optimizer — AdamW | lr 8e-4 cosine | wd 1e-4

Budget — 20k x 64 = 1.28M triples

8 ROBUSTNESS & ANALYSIS

Procedural split protocol

TRAIN
272 atlases
seeds 0-16
widths 128, 160, 224, 280

RESEARCH
256 atlases
seeds 9001-9008
8 widths: 104-376

SEALED FINAL
64 atlases
seeds 10001-10004
widths 112, 181, 299, 367

All splits span four themes. Research and final seeds and widths are absent from training. The sealed holdout is evaluated once, then retired.

DIAGNOSTICS
Failures group by theme, seed, width, action rank, and attribute. Specialist logits expose perception errors without becoming a hard gate.

Finite envelope claim - not proof of universal 100% reliability.

4 CORE ARCHITECTURE

CARD BATCH
[N, 3, 21, 30]
RGB f32

→ **SHARED CONV**
5 x 5 | stride 2 | pad 2
3 -> 12 channels | 912 params

→ **SHARED ACTIVATION**
exact-erf GELU
[N, 12, 11, 15]

↓ fan-out: same shared map enters four independent learned branches ↓

COUNT	COLOR	SHAPE	FILL
Conv2d: 12 -> 8 | k3 s2 p1 [N, 8, 6, 8]	Conv2d: 12 -> 8 | k3 s2 p1 [N, 8, 6, 8]	Conv2d: 12 -> 8 | k3 s2 p1 [N, 8, 6, 8]	Conv2d: 12 -> 8 | k3 s2 p1 [N, 8, 6, 8]
GELU ↓ PROJECTION: flatten 384 -> 24 -> 3, GELU + softmax | 10,187 params	GELU ↓ PROJECTION: flatten 384 -> 24 -> 3, GELU + softmax | 10,187 params	GELU ↓ PROJECTION: flatten 384 -> 24 -> 3, GELU + softmax | 10,187 params	GELU ↓ PROJECTION: flatten 384 -> 24 -> 3, GELU + softmax | 10,187 params

ENUMERATE
12 choose 3 = 220
candidate actions

→ **TRIPLE EVIDENCE**
for each attribute:
p(i) + p(j) + p(k)
shape [220, 4, 3]

→ **ATTRIBUTE MLP**
shared 3 -> 32 -> 1
GELU | x4
161 params

→ **SET MLP**
concat 4 -> 16 -> 1
GELU
97 params

→ **ACTION**
argmax score
-> [i, j, k]

5 ATTRIBUTE SPECIALIST OUTPUTS

COUNT
1 | 2 | 3
p = [p1, p2, p3]

COLOR
red | green | purple
p = [pr, pg, pp]

SHAPE
diamond | squiggle | oval
3 learned logits

FILL
solid | striped | outline
3 learned logits

Auxiliary labels make each branch inspectable. The relation scorer consumes probabilities, not argmax labels, so uncertainty and gradients remain continuous.

7 INFERENCE & DEPLOYMENT

BROWSER
capture + crop
area resample

→ **WASM**
FrankenTorch
embedded .ftsv

→ **ACTION**
packed u32
highlight / click

Checkpoint	330 KB .ftsv
Generated WASM	2.5 MiB
Cached inference	2.416 ms / board
With pixel transfer	2.508 ms / board

9 BENCHMARKS & EVALUATION

0 / 4,096	0 / 10,000	99.9975%	12 / 12
research board failures	sealed board failures	sealed triple accuracy	browser action paths

20k-step training time - 24.6x faster



Board inference latency



10 AUTORESEARCH LOOP & PROMOTION GATES

HYPOTHESIS
one falsifiable change

→ **TRAIN**
fixed 20k-step budget

→ **RESEARCH**
4,096 boards
keep / discard

→ **SEALED**
10,000 boards
exactly once

→ **PROMOTE**
same checkpoint
-> WASM

Selection order

1. **fewer board failures**
2. **higher triple accuracy**
3. **lower latency, then size**

Reproducibility controls

Seed	99005
Batch	64 balanced triples
Run record	no augment | append-only log

11 MODEL FACTS

Policy	compact CNN
Attention	none
Memory	none
Runtime solver	none
Output	3 card indices
Parameters	41,918
Actions	220 / board

LEGEND - MODULE COLORS

- shared CNN
- relation scorer
- runtime / state
- specialists
- loss / training
- browser action

TENSOR / SHAPE GUIDE

N	card count or batch cards
C, H, W	channels, height, width
[N, 4, 3]	four attributes x three values
[220, 4, 3]	all triple evidence vectors

GRADIENT / DATA FLOW

→ solid: inference data flow ↓ branch / supervision direction

Training loss backpropagates through both relation MLPs, probability sums, each specialist softmax and classifier, specialist convolutions, and the shared convolution. There is no nondifferentiable gate between loss and specialists.

IMPLEMENTATION SOURCES

Policy and browser path: `vision-policy/src/lib.rs`
Fast training graph: `vision-mpsgraph/src/policy_graph.rs`
Pixel contract: `extension/preprocess.js`
Protocol and results: `vision/AUTORESEARCH.md`, `vision/RESEARCH_LOG.md`

Diagram generated from editable Typst source. Metrics measured on Apple M5 Pro, 2026-08-18.